

2026 src/scx/
Target: Synthese / Philosophy of Science / Erkenntnis

SCX

State-Conditioned eXpertise and Scientific Consensus:
A Formalization of Philosophy of Science

SCX

July 2, 2026

Abstract

—— SCXState-Conditioned eXpertise $M\mathcal{S}$
 M Chernoff-Hoeffding $\Delta_s > 0$ $M \geq M_{\text{crit}}$ s $M \exp(-2M\Delta_s^2)$
 Kuhn $\mathcal{S}_K$ $\mathcal{E}_K D_{\text{new}}$ Fano —— $I(Y; X_{\text{new}} | \mathcal{S}_K) > H(Y | \mathcal{S}_K) - \log 2$ ——
 SCXPopperSCX s $(H_0 : \Delta_s \leq 0, H_1 : \Delta_s > 0)$ $\Delta_s^{\text{claimed}}$ $\mathcal{V}$ $\Delta_s^{\text{claimed}}$
 SCX $\{E'_1, \dots, E'_{M'}\}$ $\hat{\Delta}_s^{\text{rep}} \leq 0$ Chernoff SCX
Honest Critique//
 SCXChernoff-HoeffdingFano

1

1.1

20Popper KuhnLakatosFeyerabend

- (1) Cover & Thomas (2006)Wainwright (2019) Boucheron(2013)
- (2) SCXYajie(SCX Project, 2026) “”
- (3) Open Science Collaboration (2015) Ioannidis (2005)“”

SCX———
 M Δ_s Chernoff-HoeffdingFano

1.2 SCX

SCX

Definition 1.1 (——). $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ s_i

- (i) $\phi(s_i) \in \mathbb{R}^d$
- (ii) $y_i \in \mathcal{Y}$ //
- (iii) $/p_i \in \mathcal{P}$

Definition 1.2 (——). $M\mathcal{E} = \{E_1, E_2, \dots, E_M\}$ sm

$$v_m(s) = \mathbf{1}[E_m(\phi(s)) \neq y] \in \{0, 1\} \quad (1)$$

$$\mathbf{1}[\cdot]$$

$$v_m(s) = 0sv_m(s) = 1$$

“”

Axiom 1.3 (——). y_s

$$v_m(s) \perp\!\!\!\perp v_{m'}(s) \mid y_s, \quad \forall m \neq m' \quad (2)$$

$\square$ —— §6.1

Axiom 1.4 (——). $y_s v_m(s) = 1p_{clean,s} < 0.5$ $y_s v_m(s) = 1p_{noisy,s} > 0.5$

$$\boxed{\Delta_s = p_{noisy,s} - p_{clean,s}} \quad (3)$$

$$\Delta_s > 0$$

Axiom 1.5 (——Fano). *Fano*

$$P_e \geq \frac{H(Y \mid X) - 1}{\log |\mathcal{Y}|} \quad (4)$$

$$P_e H(Y \mid X)XY$$

——

2 M

2.1

——

Definition 2.1. $s \in SM\{v_1(s), \dots, v_M(s)\}$

$$C_M(s) = \frac{1}{M} \sum_{m=1}^M (1 - v_m(s)) = 1 - \frac{1}{M} \sum_{m=1}^M v_m(s) \quad (5)$$

$$s$$

$$s\alpha\text{-}\tau_\alpha \in (0.5, 1]$$

$$\boxed{\mathbb{P}(C_M(s) \geq \tau_\alpha \mid \Delta_s > 0) \geq 1 - \alpha} \quad (6)$$

$$M1 - \alpha$$

$$M\alpha\tau_\alpha$$

2.2 ——

Theorem 2.2 (Honest Person Theorem). $M\Delta_s > 0$ $M \rightarrow \infty C_M(s)11 - p_{clean,s}$

$$\boxed{\mathbb{P}\left(\lim_{M \rightarrow \infty} C_M(s) = 1 - p_{clean,s}\right) = 1} \quad (7)$$

$$\varepsilon > 0C_M(s)\varepsilon\text{Chernoff-Hoeffding}$$

$$\boxed{\mathbb{P}(|C_M(s) - (1 - p_{clean,s})| \geq \varepsilon) \leq 2 \exp(-2M\varepsilon^2)} \quad (8)$$

Proof. $\mathbf{1}y_s v_m(s) \sim \text{Bernoulli}(p_{\text{clean},s}) \xrightarrow{\text{a.s.}} p_{\text{clean},s}$ $C_M(s) = 1 - \frac{1}{M} \sum v_m(s) \xrightarrow{\text{a.s.}} 1 - p_{\text{clean},s}$

2Chernoff-Hoeffding $C_M(s)M[0, 1]$ Hoeffding $\varepsilon > 0$

$$\mathbb{P}(|C_M(s) - \mathbb{E}[C_M(s)]| \geq \varepsilon) \leq 2 \exp(-2M\varepsilon^2) \quad (9)$$

$$\mathbb{E}[C_M(s)] = 1 - p_{\text{clean},s}$$

32

$$\tau_\alpha = 1 - p_{\text{clean},s} - \sqrt{\frac{\log(2/\alpha)}{2M}} \quad (10)$$

$$\mathbb{P}(C_M(s) \geq \tau_\alpha) \geq 1 - \alpha$$

□

□

□ 12Hoeffding(1963) 3????

“” ——— Condorcet

2.3 ———

“” Galileo, Wegener, Boltzmann

Theorem 2.3 (Unlikely-to-be-Solo-Awake Theorem). $M - 1 \Delta_s > 0$

$$\mathbb{P}\left(|C_M(s) - \frac{M-1}{M}, \Delta_s > 0\right) \leq \frac{p_{\text{noisy},s}}{p_{\text{noisy},s} + (M-1)(1-p_{\text{clean},s}) \cdot \frac{p_{\text{noisy},s}^{M-2}}{p_{\text{clean},s}^{M-2}} \cdot \frac{1-p_{\text{noisy},s}}{1-p_{\text{clean},s}}} \quad (11)$$

$$M \rightarrow \infty \exp(-2M\Delta_s^2)$$

Proof. **1Bayes** $H_0 s H_1 s$ $M - 1$

$$\mathbb{P}(H_1 | \text{data}) = \frac{\mathbb{P}(\text{data} | H_1) \cdot \mathbb{P}(H_1)}{\mathbb{P}(\text{data} | H_0) \cdot \mathbb{P}(H_0) + \mathbb{P}(\text{data} | H_1) \cdot \mathbb{P}(H_1)} \quad (12)$$

$$\mathbf{2}H_0 \mathbb{P}(\text{data} | H_0) = p_{\text{clean},s}^{M-1} \cdot (1 - p_{\text{clean},s})M - 1 \quad H_1 \mathbb{P}(\text{data} | H_1) = p_{\text{noisy},s} \cdot (1 - p_{\text{noisy},s})^{M-1}M - 1$$

3odds

$$\frac{\mathbb{P}(H_1 | \text{data})}{\mathbb{P}(H_0 | \text{data})} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \cdot \frac{p_{\text{noisy},s}}{1 - p_{\text{clean},s}} \cdot \left(\frac{1 - p_{\text{noisy},s}}{p_{\text{clean},s}}\right)^{M-1} \quad (13)$$

$$p_{\text{clean},s} < 0.5 < p_{\text{noisy},s} \Delta_s > 0 \quad 1 - p_{\text{noisy},s} < 0.5 < p_{\text{clean},s} \quad (1 - p_{\text{noisy},s})/p_{\text{clean},s} < 1M$$

4Chernoff Hoeffding

$$\mathbb{P}(H_1 | \text{data}) \leq \exp(-2M\Delta_s^2 + O(1)) \quad (14)$$

$$\Delta_s = p_{\text{noisy},s} - p_{\text{clean},s} > 0$$

□

□

□ 1-23 $\Delta_s > 0$ 4ChernoffHoeffding $\mathbb{P}(H_0) \approx \mathbb{P}(H_1)$ $H_1??$

Remark 2.4. (i) $\mathbb{P}(H_1)$ (ii) (i)p-hacking ——— M (ii) $M_{\text{eff}} < M$ §6.1

2.4 Chernoff

Theorem 2.5 (—— Chernoff). $sM\alpha$ s

$$C_M(s) \geq \frac{1}{2} + \sqrt{\frac{\log(1/\alpha)}{2M}} \quad (15)$$

$$M \geq M_{\text{crit}} = \frac{\log(1/\alpha)}{2(C_M(s) - 1/2)^2} \quad (16)$$

Proof. $H_0 : \Delta_s \leq 0$ $C_M(s) \leq 1/2$ Hoeffding

$$\mathbb{P}_{H_0} \left(C_M(s) \geq \frac{1}{2} + t \right) \leq \exp(-2Mt^2) \quad (17)$$

$$= \alpha t = \sqrt{\log(1/\alpha)/(2M)} H_0 1/2 + \sqrt{\log(1/\alpha)/(2M)} M M_{\text{crit}} \quad \square \quad \square$$

$$\begin{aligned} \square \quad ?? \quad \Pi \Delta_s \beta &\approx \exp(-2M(\Delta_s - t)^2) \Delta_s 0M \\ \Delta_s &\text{---} \S 3 \end{aligned}$$

3 KuhnFano

3.1 -

Thomas Kuhn (1962) “” “” Kuhn—— SCX Fano

Definition 3.1 (KuhnSCX). *Kuhn* $\mathcal{K}$

$$\boxed{\mathcal{K} = (\mathcal{S}_K, \mathcal{E}_K)} \quad (18)$$

- $\mathcal{S}_K \subset \mathcal{S}$ — “”
- $\mathcal{E}_K = \{E_1^K, \dots, E_{M_K}^K\}$ —

$$H_K(Y \mid \mathcal{S}_K) = H(Y) - I_K(Y; \mathcal{S}_K) \quad (19)$$

$$I_K(Y; \mathcal{S}_K) \mathcal{K}$$

Definition 3.2 (—). $\mathcal{K} D_{\text{new}} = \{(x_i, y_i)\}_{i=1}^n \varepsilon$ -

$$\boxed{I(Y; D_{\text{new}} \mid \mathcal{S}_K) > \varepsilon} \quad (20)$$

ε

3.2 Fano

Theorem 3.3 (—Fano). $\mathcal{K} D_{\text{new}}$ Fano

$$\boxed{I(Y; D_{\text{new}} \mid \mathcal{S}_K) > H(Y \mid \mathcal{S}_K) - \log 2} \quad (21)$$

$$\mathcal{K} \mathcal{E}_K 1/2 \text{---}$$

Proof. **1Fano** $\mathcal{K}$

$$P_e^{\mathcal{K}} \geq \frac{H(Y \mid \mathcal{S}_K) - 1}{\log |\mathcal{Y}|} \quad (22)$$

$$\begin{aligned} /|\mathcal{Y}| &= 2 \log |\mathcal{Y}| = \log 2 \\ \mathbf{2} D_{\text{new}} \end{aligned}$$

$$H(Y \mid \mathcal{S}_K, D_{\text{new}}) = H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) \quad (23)$$

3Fano

$$P_e^{\mathcal{K} + D_{\text{new}}} \geq \frac{H(Y \mid \mathcal{S}_K, D_{\text{new}}) - 1}{\log 2} = \frac{H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1}{\log 2} \quad (24)$$

$$4P_e^{\mathcal{K}+D_{\text{new}}} > 1/2$$

$$\frac{H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1}{\log 2} > \frac{1}{2} \quad (25)$$

$$H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1 > \frac{1}{2} \log 2 \quad (26)$$

$$H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) > 1 + \frac{1}{2} \log 2 \quad (27)$$

nat

$$I(Y; D_{\text{new}} \mid \mathcal{S}_K) > H(Y \mid \mathcal{S}_K) - \log 2 \quad (28)$$

□

□

□ Fano(Cover & Thomas, 2006, 2.10.1) 24
Kuhn“”

3.3

Theorem 3.4. $\mathcal{K}_{old}\mathcal{K}_{new}$

$$\boxed{G(\mathcal{K}_{new} \mid \mathcal{K}_{old}) = I(Y; \mathcal{S}_{new}) - I(Y; \mathcal{S}_{old})} \quad (29)$$

$r(t)$ ————

$$\boxed{r(t) \propto \frac{G(\mathcal{K}_{new} \mid \mathcal{K}_{old})}{H(\mathcal{S}_{new} \mid \mathcal{S}_{old})} \cdot \frac{1}{\tau_{gen}}} \quad (30)$$

τ_{gen} “” $H(\mathcal{S}_{new} \mid \mathcal{S}_{old})$

. $\mathbf{1}E_m \ U(E_m; \mathcal{K}) \ G \rightarrow$
 $\mathbf{2}H(\mathcal{S}_{new} \mid \mathcal{S}_{old})$ “” $\rightarrow$
 $\mathbf{3}$ Kuhn——“ ”—— τ_{gen} /
 $\square \ G/H1/\tau_{gen}$

□

Corollary 3.5 (Kuhn——Planck). **Planck**“ ”SCX

$$\boxed{t_{shift} = \tau_{gen} \cdot \frac{H(\mathcal{S}_{new} \mid \mathcal{S}_{old})}{G(\mathcal{K}_{new} \mid \mathcal{K}_{old})}} \quad (31)$$

$t_{shift} \ t_{shift} \gg \tau_{gen}$ ——

□ (i) $G(\mathcal{K}_{new})$ (ii) $H(\mathcal{S}_{new} \mid \mathcal{S}_{old})$ Kuhn

4 PopperSCX

4.1

Karl Popper (1935/1959) Popper

(1) **Duhem-Quine**——

(2) —— “> 70%”

SCX

4.2 SCX

Definition 4.1 (SCX—). $s \in \mathcal{SSCX}$ -

$$\boxed{\mathcal{F}(s) = (\Delta_s^{\text{claimed}}, \mathcal{V}, M_{\min})} \quad (32)$$

- $\Delta_s^{\text{claimed}} > 0$ — M
- $\mathcal{V}$ —
- $M_{\min}$ — $\text{Chernoff} H_0 H_1$

Theorem 4.2 (—SCXPopper). $s\text{SCX-}\mathcal{VM}$ power $> 1/2$

$$H_0 : \Delta_s \leq 0 \quad \text{—} \quad (33)$$

$$H_1 : \Delta_s \geq \Delta_s^{\text{claimed}} > 0 \quad \text{—} \quad (34)$$

$\Delta_s^{\text{claimed}} \mathcal{V}$ $s\text{SCX}$ - —

Proof. $1\mathcal{F}(s)$

- $\Delta_s^{\text{claimed}} \rightarrow / \rightarrow$
- $\mathcal{V} \rightarrow \rightarrow$
- $M_{\min} \rightarrow \text{—} \text{“”}$

2—Chernoff $\mathcal{F}(s)$ Chernoff

$$\text{Power}(M, \Delta_s^{\text{claimed}}, \alpha) = \mathbb{P}_{H_1} \left(C_M(s) \geq \frac{1}{2} + \sqrt{\frac{\log(1/\alpha)}{2M}} \right) \geq 1 - \exp \left(-2M \left(\Delta_s^{\text{claimed}} - \sqrt{\frac{\log(1/\alpha)}{2M}} \right)^2 \right) \quad (35)$$

$M \geq M_{\min} M_{\min} > 1/2$

3Duhem-Quine $\text{SCX} A_1, \dots, A_k \text{SCX } \phi(s) \mathcal{V} H_0 \text{SCX} \text{“”} \text{—Duhem-Quine}$

4SCX “ $p > 0.7$ ” $\Delta_s \text{Bernoulli } \Delta_s = p - (1 - p) = 2p - 1p > 0.5$ $\square$ $\square$

$\square$ Hoeffding Duhem-Quine $\text{“”} I(Y; A_j \mid S, \{A_k\}_{k \neq j}) \square$

4.3

Theorem 4.3 (SCX). SCX

(**UF1**) $\Delta_s^{\text{claimed}} \text{—} \text{“”} \Delta_s \rightarrow \text{Chernoff}$

(**UF2**) $\mathcal{V} \text{—} \text{“”} \rightarrow$

(**UF3**) $M_{\min} \text{—} \text{“”} \lim_{M \rightarrow \infty}$

$s\text{SCX}$ -

Proof. $??(1)\mathcal{F}(s)$ $\square$ $\square$

$\square$ **SCX** SCX

(1) $\mathcal{V}$ —

(2) $\Delta_s^{\text{claimed}} \Delta_s^{\text{claimed}} M$

(3) $\mathcal{S}$ — Kuhn-Feyerabend SCX

SCX SCX

5 SCX

5.1 SCX

replication crisis Open Science Collaboration (2015)36% Camerer(2018)62%Errington(2021) 46%
SCX

Definition 5.1 (—SCX). $s\mathcal{E}_{orig} = \{E_1, \dots, E_M\}$ $\mathcal{E}_{rep} = \{E'_1, \dots, E'_{M'}\}$ $\forall SCX$

$$\boxed{\hat{\Delta}_s^{rep} > 0 \quad \left| \hat{\Delta}_s^{rep} - \hat{\Delta}_s^{orig} \right| \leq \sqrt{\frac{\log(2/\alpha)}{2} \left(\frac{1}{M} + \frac{1}{M'} \right)}} \quad (36)$$

$$\hat{\Delta}_s^{orig} \hat{\Delta}_s^{rep}$$

5.2 SCX

Theorem 5.2. *SCX*

1 Margin Reversal $\hat{\Delta}_s^{rep} \leq 0$ $\hat{\Delta}_s^{orig} > 0$

2 Expert Collapse $\hat{\Delta}_s^{rep} \approx \hat{\Delta}_s^{orig} > 0$ $M'_{eff} \ll M'_{nominal}$ —

3 State Occlusion $\hat{\Delta}_s^{rep}(\mathcal{S}') > 0$ $\hat{\Delta}_s^{rep}(\mathcal{S}) \leq 0$ —

Proof. **1??** $\hat{\Delta}_s^{rep} > 0$ $C_{M'}(s)1/2$ Chernoff?? Δ_s p-hacking

2 $\mathcal{E}_{rep}$ $M'_{eff} < M'_{nominal}$ Chernoff-HoeffdingJanson, 2004 $\hat{\Delta}_s^{rep} O(1/M'_{eff})O(1/M')$ $M'_{eff} \ll M'_{nominal}$
“” “”

3 $\mathcal{S}$ $\mathcal{S}' \subset \mathcal{S}$ $\mathcal{S} \setminus \mathcal{S}' \rightarrow$

$$\hat{\Delta}_s^{rep}(\mathcal{S}) = \frac{|\mathcal{S}'|}{|\mathcal{S}|} \hat{\Delta}_s^{rep}(\mathcal{S}') + \frac{|\mathcal{S} \setminus \mathcal{S}'|}{|\mathcal{S}|} \hat{\Delta}_s^{rep}(\mathcal{S} \setminus \mathcal{S}') \quad (37)$$

$$\hat{\Delta}_s^{rep}(\mathcal{S} \setminus \mathcal{S}') \leq 0 \quad |\mathcal{S}'|/|\mathcal{S}| \quad \text{“X”X} \quad \square \quad \square$$

$\square$ $\hat{\Delta}_s$ Hoeffding

5.3

Theorem 5.3.

$$\boxed{I(Y; \mathcal{E}_{orig}) + I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig}) \leq H(Y)} \quad (38)$$

$$I(Y; \mathcal{E}_{orig}) \quad I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig}) \quad \text{—}$$

Proof.

$$I(Y; \mathcal{E}_{orig}, \mathcal{E}_{rep}) = I(Y; \mathcal{E}_{orig}) + I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig}) \leq H(Y) \quad (39)$$

$$I(Y; \mathcal{E}_{orig}) \approx H(Y) \quad \text{“”} \quad I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig}) \approx 0 \quad \text{—} \quad \square \quad \square$$

$\square$?? —

M “” — “”NSCX

Corollary 5.4 (SCX).

$$\boxed{I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig})} \quad (40)$$

—

6

6.1

??

- (1) $\rightarrow$
- (2) **Monoculture** $R \rightarrow$ “”
- (3) $\rightarrow$ “”
- (4) Kuhn $\rightarrow \Delta_s$

Theorem 6.1 (——). *Chernoff-Hoeffding* M M_{eff}

$$\boxed{M_{\text{eff}} = \frac{M}{1 + (M - 1)\bar{\rho}}} \quad (41)$$

$\bar{\rho} ICC$ $\bar{\rho} \rightarrow 1$ $M_{\text{eff}} \rightarrow 1$ ——

Proof. $M\{v_1, \dots, v_M\} \text{Var}(v_m) = \sigma^2$ $\text{Cov}(v_m, v_{m'}) = \rho\sigma^2 m \neq m'$ $\bar{v} = \frac{1}{M} \sum v_m$

$$\text{Var}(\bar{v}) = \frac{\sigma^2}{M} + \frac{M-1}{M} \rho\sigma^2 = \frac{\sigma^2}{M} (1 + (M-1)\rho) \quad (42)$$

$\rho = 0$ $M/(1 + (M-1)\bar{\rho})$ Hoeffding □ □

□ M_{eff}
 $M_{\text{eff}} M_{\text{nominal}} 10\% - 30\%$ “ $M = 100$ ” 10–30

6.2 ——— Gödel-Turing

Remark 6.2 (Gödelian). $SCX??$ $SCX\Delta_s^{\text{claimed}}$ $\mathcal{V}SCX-$

- (1) SCX “ SCX ” SCX
- (2) SCX
- (3) SCX —— $SCXSCX$

Gödel

□ SCX $SCXSCX??-??$ SCX —— Gödel-Turing

7 ——— SCX

7.1

- (1) §2 “” ——
- (2) §3 —— Fano
- (3) §4 —— “”
- (4) §5 ——

$SCX\Delta_s$

Table 1: SCX

		SCX
Popper		$\Delta_s^{\text{claimed}} \mathcal{V} \text{ ??}$
Kuhn		$= \text{Fano} \text{ ??}$
Lakatos	++	$\mathcal{S}_K + \text{??}3$
Condorcet		??
Feyerabend	“”	$M_{\text{eff}} \rightarrow 1$ —— “”??
IOANNIDIS	“”	??
(2005)		

7.2

7.3

??

—— $\bar{\rho} \, p_{\text{clean},s}$
?? (i)
(ii) $\exp(-2M\Delta_s^2)$ —— Δ_s

7.4

—— ——SCX
□□ □

- (1) Gödel??
- (2) ??
- (3) ?? $H(\mathcal{S}_{\text{new}} \mid \mathcal{S}_{\text{old}})$ ——

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SCX(SCX Project, 2026) “” “”SCX SCX—— “”

References

[1] S. Boucheron, G. Lugosi, and P. Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[2] C. F. Camerer *et al.* Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2:637–644, 2018.

[3] T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd edition. Wiley, 2006.

[4] T. M. Errington *et al.* Investigating the replicability of preclinical cancer biology. *eLife*, 10:e71601, 2021.

[5] P. Feyerabend. *Against Method*. New Left Books, 1975.

- [6] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.
- [7] J. P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005.
- [8] S. Janson. Large deviations for sums of partly dependent random variables. *Random Structures & Algorithms*, 24(3):234–248, 2004.
- [9] T. S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, 1962.
- [10] I. Lakatos. *The Methodology of Scientific Research Programmes*. Cambridge University Press, 1978.
- [11] Open Science Collaboration. Estimating the reproducibility of psychological science. *Science*, 349(6251):aac4716, 2015.
- [12] K. R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959. (Original work published 1935)
- [13] W. V. O. Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1):20–43, 1951.
- [14] SCX Project. Theorem 1–4: Multi-expert consistency, weak feature limits, unidentifiability, and minimax optimality. Technical report, `src/scx/`, 2026.
- [15] SCX Project. Situs: Physics-anchored positional encoding for state-conditioned expertise. Preprint, `paper/situs_theory/main.tex`, 2026.
- [16] SCX Project. The Yajie Protocol: Technology lock-in, audit sovereignty, and the non-proliferation logic of data quality assessment. Working paper, `yajie_protocol_paper.md`, 2026.
- [17] M. J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press, 2019.